Sample-free Synthetic Population Generation for Agent-Based Models

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Abstract

Generating synthetic populations of agents that conform to the data available on real populations has recently become a necessity for social modelers, and although several approaches have been proposed in recent works, it still constitutes a significant scientific and methodological challenge because most of them rely on individual data gathered from samples of the population. We will describe in this talk an alternative approach based on the use of population-wide statistical data and a software tool that takes advantage of this approach to generate realistic synthetic populations in agent-based models.

Keywords: Synthetic Population Generation, Agent-Based Modeling

1. Synthetic populations in agent-based models

Agent-based social simulation is gaining ground as a candidate of choice for building decision-support tools in the management of complex socio-environmental systems, and the resulting models are being designed to produce more realistic outcomes. Generating synthetic populations of agents that conform to the data available on real populations therefore becomes a critical necessity for social modelers, and although several approaches have been proposed in recent works, it still constitutes a significant scientific and methodological challenge as most of them require individual data gathered from samples of the population.

2. Sample-based and sample-free approaches

Acquiring data from population samples (through surveys, for instance) is costly, not always feasible, and potentially made obsolete by the growing public availability of various demographical data (from censuses or initiatives like Open Data). There are, however, very few methods (known as « sample-free » methods) that can exploit these global datasets to produce synthetic populations.

3. Sample-free population generator

We introduce in this talk a tool that can be fed with statistical data about a population and generate synthetic populations of any size with a distribution of attributes among agents equivalent to the ones observed in reality among human beings. It relies on a meta-model that allows to specify, firstly, the variables of interest in the input data (e.g., sex, income, occupation, etc.) and their relationships with the attributes of the agents, and, secondly, a target agent-based modeling and simulation platform (e.g., GAMA [4]) in which it can initialise a model with the generated populations ready to be simulated.

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